

Class #2 - Matrices

1. Size of a matrix: (row) $\times$ (column)

2. Def: Two matrices are called *row equivalent* if there is a sequence of row operations which transforms one matrix to the other.

Examples: Fill in the blanks.

- (a) The _____ matrices associated with (I)-(IV) are _____ equivalent.
- (b) The _____ matrices associated with (I)-(IV) are _____ equivalent.
- (c) The Augmented matrices associated with (I)-(IV) have the same _____ set.
- (d) Two _____ equivalent _____ matrices are _____.

3. Solution set of a linear system

Def: A system of linear equations is called *consistent* if it has a solution.

A system of linear equations is called _____ if it has *no* solution.

Example:
$$\begin{cases} a_{11}x_1 + a_{12}x_2 &= b_1 \\ a_{21}x_1 + a_{22}x_2 &= b_2 \end{cases}$$

4. (To be verified later) A system of linear equations has either

- (i) no solution,
- (ii) exactly one solution, or
- (iii) infinitely many solutions

5. Example: Find h so that the system is consistent and find the solution set.

$$\begin{cases} x_1 - 3x_2 &= h \\ -2x_1 + 6x_2 &= -5 \end{cases}$$

6. Def: In a row of a matrix, the *leading entry* is the first nonzero entry in that row.

Examples:

$$\begin{bmatrix} 2 & 0 & 0 & 3 \\ 0 & 0 & -3 & 1 \end{bmatrix} \qquad \begin{bmatrix} 2 & 0 & 0 & 3 \\ 1 & -1 & 1 & 1 \end{bmatrix}$$

7. Def: We say that a row of a matrix is nonzero if _____ entry in that row is nonzero.

Example:

$$\begin{bmatrix} 2 & 0 & 0 & 3 \\ 0 & 0 & -3 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

8. Def: A matrix is in *echelon* (=”step-like”) form if

- (i) leading entries form steps down and right, and
- (ii) any nonzero rows are above the zero rows.

Examples:

$$\begin{array}{ll} \text{(I)} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \text{(II)} \begin{bmatrix} 2 & 0 & 0 & 3 \\ -4 & 0 & -3 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ \text{(III)} \begin{bmatrix} 2 & 0 & 0 & 3 \\ 0 & 0 & -3 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} & \text{(IV)} \begin{bmatrix} 0 & 2 & 0 \\ 1 & 0 & -3 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

9. Def: A matrix is in *reduced echelon form* if

- (i) the matrix is in echelon form, and
- (ii) every leading entry is 1 and it is the only nonzero entry in that column.

Example:

10. Fact: Any matrix can be reduced to an echelon form using elementary row operations.

11. Note: For a matrix, there can be _____ echelon forms of that matrix.
However, there is _____ reduced echelon form of that matrix.